Chapter

Perspective Chapter: Optical Waveguides for Quantum Computation

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Abstract

Quantum computation demands various types of optical devices such as beam splitters, ring resonators, delay lines, switches, modulators, multiplexers, Mach-Zehnder interferometers, and phase shifters, to name a few. These devices are essentially engineered optical waveguides, in which the geometrical and material properties are chosen such that the optical losses are minimized, to preserve the single photon's operation. Scattering loss, arising from the wall roughness, is the dominant loss mechanism in the optical waveguides, and its minimization should be taken into account, by either design or fabrication considerations. Different types of waveguide geometries including slab waveguides, strip waveguides, rib waveguides, Bragg grating waveguides, and hybrid waveguides have been utilized in this regard. Moreover, there are three main material categories, with generic fabrication processes, regarding the refractive index contrast of the core and cover. These include low, medium, and high index contrast materials, each owing its pros and cons. Finally, designing the bend waveguides with low bending loss is highly desirable, and circular bends, sinusoidal bends, Euler bends (normal and modified), and spline bends are some of the frequently used curvatures. This chapter reviews the key points in the design of optical waveguides for quantum applications.

Keywords: integrated photonics, index contrast, optical waveguide, quantum computation, bending loss, Euler bend

1. Introduction

Quantum calculation has been the subject of intensive research recently. In this technology, a single photon is exploited as the excitation source. Thus, it is essential to carefully control potential losses in the devices to reduce the probability of losing the single photon. Importantly, losses are the main source of error in quantum computation [1]. The total loss of a planar waveguide is the sum of the contributions of material absorption, Rayleigh scattering, interfacial scattering, substrate leakage, bend radiation, and cross-talk. The material losses in the photonic integrated circuits mainly originate from the bond resonances involving hydrogen atoms and can be attacked by high-temperature annealing. Also, the leakage losses can be prevented by the deposition of thick substrates, at the expense of fabrication time. The cross-talk

can be attacked by considering sufficient distance between the waveguide cores. The interfacial losses change quadratically with the refractive index contrast and can be controlled either by material selection or fabrication processes or by introducing some specific processes, such as an oxidation-based etchless process, chemical oxidation, anisotropic etching, and post-processing [2, 3].

In classic computation, the processing speed is proportional to the number of transistors. According to the widespread rendering of Moor's law, the number of transistors on a single integrated circuit chip doubles every 18 months and the gradual miniaturization of electrical circuits is limited by quantum mechanical effects. Quantum computation, on the other hand, utilizes the superposition phenomenon, where the state of the quantum bits (qubits) is the superposition of the zero and one states. This feature increases the processing speed dramatically, since in many-qubit systems, the superposition state increases with the number of qubits [4, 5]. The electronic devices perform based on fermions (electrons), which obey Fermi–Dirac statistics, whereas, on the photonic devices, bosons (photons) are employed, obeying Bose–Einstein statistics [6].

Regarding the technology, quantum computation can be performed by superconducting qubits that demand dilution refrigerators to maintain their temperatures sufficiently low [7]. Moreover, the bulk and low-loss fiber optic system, as the name implies, is bulky, and reaching a stable performance is challenging [2, 8]. The integrated optic is highly desirable because of substantial upgradation in performance, reliability, and SWaP-C (size, weight, power, and cost). This is due to the integration of components, reduced coupling losses, and reduced sensitivity to environmental perturbation [9]. The photonic integrated circuits rely on processing the light confined to the waveguides [10]. In this technology, the quantum information is essentially the quantum states of light propagating in the photonic integrated circuit [11].

Integrated photonic platforms are formed by three main parts: a single-photon source (SPS), a circuit, and a single-photon detector (SPD). These sections are respectively utilized for generating, manipulating, and detecting the light on a small chip. Parametric photon-pair source and quantum dot (QD) single-photon source are two instances of SPSs. Moreover, avalanche photodiodes, superconducting nanowire single-photon detectors (SNSPDs), and transition edge sensors (TESs) are some of the SPD technologies [12, 13]. Similar to the classic gates, there are basic quantum gates including X, Y, Z, and Hadamard. Quantum gates are represented by unitary matrices, and each mathematical calculation can be separated into multiple such unitary calculations [14].

The ability to control the light propagation in the device after fabrication is a great demand in quantum calculation. The reconfigurable performance of the waveguides is implemented *via* (1) temperature control, (2) applying mechanical strain, (3) applying a magnetic field, (4) applying an electric field to lithium niobate ($LiNbO_3$) and 2-D materials, and (5) free-carrier plasma effect [15–18]. These methods modify the local refractive index of the waveguide modes, thus providing a phase shift in the performance. The tuning capability of the devices can be used to compensate for the impact of the fabrication imperfections in the spectral response as well [19].

Among the aforementioned methods, the thermo-optic effect offers a large dynamic range, moderate power consumption, negligible optical loss, and moderate operating speed (a few to tens of microseconds reconfiguration time) [20]. The challenge of the thermo-optic effect is the difficulty of local temperature rise in a tiny structure, which can be obviated by injecting charge carriers, providing much faster tuning time than thermal tuning method in the order of nanosecond [16, 21]. The graphene-based heaters may provide higher response speeds than those of the metallic ones [22].

2. Straight and bend waveguide design

This section discusses the various straight waveguide geometries and the key points for designing each configuration. The geometrical dimension selection and the choice of the platform play a crucial role in this regard. To extend the waveguide design to an integrated device design, it is essential to use engineered curved waveguides for connecting different sections. The main goal of the bend waveguide design is the reduction of the losses and curvatures with different mathematical formulas can be used. These materials are illustrated in the second part of this section.

2.1 Straight waveguide geometries

Integrated photonics relies on the design of simple waveguides, where other components are essentially built by introducing geometrical or physical modification to the base waveguide [23]. The performance of the waveguides is mainly evaluated by their propagation constant, attenuation constant, and insertion loss. Moreover, the transverse profile of the dielectric constant of the waveguide is employed to categorize the modes. Waveguide modes are denoted by transverse electric (TE) and transverse magnetic (TM). Moreover, degenerate modes share the same value of propagation constant but with distinguishable electric field distributions [24].

Different types of full-wave solvers have been used to simulate the performance of the optical waveguides, including but not limited to Ansys Lumerical, Comsol Multiphysics, FIMMWAVE, Rsoft, and CST commercial packages [25–29]. These solvers are based on different numerical techniques, including the vectorial beam propagation method (BPM), the finite-difference time-domain (FDTD) method, the eigenmode expansion method (EME), and the finite-element mode propagation analysis (FE-MPA). For the initial design, usually 2D or 2.5D solvers are employed. The 3D implementation of the numerical methods leads to more accurate results at the expense of more time and memory requirements. Furthermore, the 3D calculation demands considering multiple parameters such as port widths and locations [30, 31].

The simplest form of the optical waveguide is the slab waveguide, which is a thin film core, with the refractive index n_1, sandwiched between upper and lower cladding, as shown in **Figure 1a**. The refractive index of the core is higher than those of the upper and lower claddings, respectively denoted by n_2 and n_3. This structure performs based on total internal reflection, and it shows a confined wave field in the core region and exponentially decaying wave in the clad. This boundary value problem has a closed-form solution, and the effective refractive index of each propagating mode lies between those of the core and cladding. If the effective mode index of the core is less than that of the clad, the wave in the clad becomes oscillatory in the transverse direction and dissipates as the radiating mode [33]. In this structure, the light is confined in only one direction, and high-power waveguide lasers and amplifiers benefit from this geometry [24].

The three-dimensional version of the slab waveguides is of the strip type, illustrated in **Figure 1b**. It is constructed by a high-index core on top of the substrate. The effective refractive indexes of these layers are denoted by n_1 and n_2. To reduce the leakage from the substrate, the core can be covered by a superstrate and form a waveguide, the so-called buried channel waveguide, as shown in **Figure 1c**. The width w and height d of the waveguide can be manipulated to excite the propagation of the desired modes. The three-dimensional waveguide can be formed by either a rectangular or cylindrical core, supported by a substrate and/or superstrate.

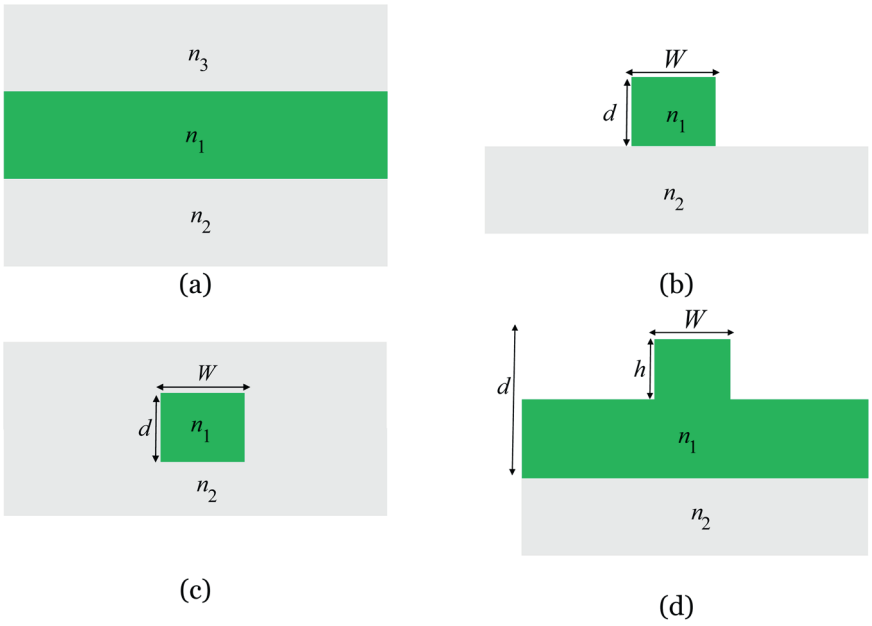


Figure 1.
The cross section of different waveguide geometries: (a) slab waveguide, (b) strip (ridge) waveguide, (c) buried channel waveguide, and (d) rib waveguide ($n_1 > n_2 > n_3$). The three-dimensional views of the waveguides can be found in Ref. [32].

The strip waveguide results in light confinement in two dimensions; thus, it is suitable for applications where diffraction-less performance is required. The device can be modeled by semi-analytical or numerical calculation since there is not any general exact analytical solution for this geometry and under specific conditions, approximate solutions can be attained [10]. Effective-index method, Marcatili's method, the mode-matching method, the finite-element method, the finite-difference method, the weighted-index method, and the perturbation theory are some of the implemented numerical tools for the analysis of these waveguides [34]. This geometry can be considered as the main building block for the optical device design.

To realize the propagating modes in optical waveguides, the mode dispersion diagram is used. This diagram shows the supported modes in the waveguide as a function of the waveguide width, and it is an essential tool for waveguide design [35]. A typical mode dispersion diagram is shown in **Figure 2a**, from which the onset of the propagation of each transverse mode can be recognized. Thus, it is feasible to propagate the desired number of modes in the waveguides by the proper choice of the core width for a given substrate. The electric field distribution for each transverse mode is shown in **Figure 2b** [36]. The anti-crossing between the TE and TM modes in the mode dispersion diagram is the evidence of hybridization, which can be further confirmed by the polarization ratio [37]. Such hybridization is not visible in the provided figure.

To have some rule-of-thumb design guidelines for the waveguide height selection, let us review its impact on the waveguide performance. For the rectangular cores, as the height of the substrate reduces, the mode confinement also reduces; thus, the modes can penetrate deeper into the cladding. On the other hand, the ultrathin waveguides have less amount of losses due to the sidewall roughness and they exhibit more robust performance against fabrication imperfections [38]. This is related to

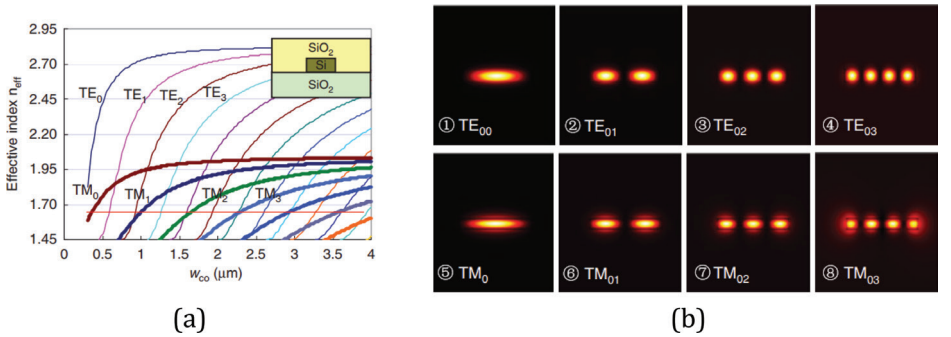


Figure 2.
(a) The mode dispersion of a typical waveguide. (b) Transverse electric field distribution in different modes [36].

the fact that the roughness of the top and bottom waveguide walls can be removed by polishing; thus, the scattering losses mainly arise from sidewalls [39]. In general, thin waveguides operate at the single dominant TE mode and they exhibit good compromise in compactness (due to the lower bending radius), propagation loss, and detrimental nonlinear effect [40]. By using the 220 nm silicon on insulator (SOI) platform, the standard complementary metal-oxide-semiconductor (CMOS) process can be used for the fabrication [41]. The strain issues may be considered when choosing the layer thickness [42].

Given the waveguide aspect ratio, TE modes are more confined than TM modes and exhibit a larger effective index. Thus, in the latter case, a non-negligible part of the wave propagates in the cladding due to the delocalized nature of the TM wave fields. Note that the TE and TM waves are respectively laterally and vertically confined [43]. Birefringence, the difference of the TE and TM effective refractive indexes, is typically used to characterize the polarization sensitivity of the wave propagation in waveguides [44]. Another parameter that can aid in quantifying the localizations is the effective area [45]. Furthermore, the group index n_g of the waveguide modes can be calculated using the associated mode effective index n_{eff} as [46]:

$$n_g = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda} \tag{1}$$

where λ is the wavelength and $dn_{eff} / d\lambda$ the first derivative of the effective index with respect to wavelength.

Rib waveguide technology (**Figure 1d**) is another promising geometry for integrated photonic devices. In this configuration, the etching depth can be used to control the scattering loss due to the side wall roughness. In this regard, rib waveguides with single and double etching steps can be utilized. By making the sidewall boundary of the bottom etch in the former case far away from the modal field, it may have little influence on modal fields [47]. Moreover, to meet the single-mode propagation condition, the height ratio of the rib waveguide might satisfy a specific condition, that is, to be larger than 0.5 in the SOI technology. The height ratio is defined as the height of the base waveguide to the total height. Also, by designing rib waveguides with large cross sections, fiber-waveguide coupling loss can be reduced [48]. The etching depth can also be used to suppress the higher-order modes by differential leakage loss [49].

As another category, subwavelength gratings (SWGs) are periodic or aperiodic geometries, where diffraction effects are suppressed by using a subwavelength

grating pitch (Λ). SWGs have been employed in planar waveguides for fiber-chip surface grating couplers, microphotonic waveguides, lenses, waveguide crossings, fiber-chip edge couplers, wavelength multiplexers, ultrafast optical switches, dispersion engineering in integrated optics, and colorless directional couplers [50]. **Figure 3c** and **d** illustrates different Bragg grating waveguides with the periodic spatial modulations respectively in waveguide height, waveguide width, slab width, and cladding [51]. Bragg grating waveguides are commonly designed for TE mode operation. By using the TM modes in the design, various advantages including high quality and smoother spectral response, low propagation loss, negative group delay slope for dispersion compensation, and low coupling coefficient can be achieved [52].

Augmented low-index guiding (ALIG) waveguides consist of a low-index substrate (silica), a low-index cladding layer (air or silica), and a two-layer core made up of two dielectric materials with a high-index contrast (silicon and silicon nitride). This waveguide is based on a silicon nitride-enhanced silicon-on-insulator platform, where the TM mode is mostly confined in the low-index silicon nitride, while the TE mode is confined in the high-index silicon [53].

2.2 Waveguide design platforms

Before proceeding to bend waveguide geometries, let us review the common materials incorporated in the photonic integrated technology and discuss their advantages and disadvantages. There are several platforms for photonic integrated circuit design including silicon-on-insulator (SOI), III-V semiconductors, silicon nitride (Si_3N_4), lithium niobate ($LiNbO_3$), polymers, and rare-earth-ion (RE^{3+})-doped materials. Among all, the SOI platform, the InP platform, and the Si_3N_4 platform offer generic fabrication processes. They are commercially accessible *via* multi-project wafer (MPW) runs [54]. An ideal platform possibly combines the efficiency of electrically pumped optical sources available in GaAs and InP systems, with high-contrast Si waveguide structures or ultralow-loss waveguides available in the Si_3N_4 platform [9]. Also, because of its strong electro-optical coefficient, lithium niobate on insulator (LNOI) is another promising PIC platform to increase single-channel data transmission speed. This CMOS-compatible platform supports both monolithic and hybrid integration [55].

Silicon nitride (Si_3N_4) is gradually becoming an attractive dielectric waveguide material due to its low propagation loss, wide transparent window for wavelength from 0.4 μm up to 2.35 μm, CMOS-compatible fabrication, and large bandgap of about 5 eV resulting in negligible two-photon absorption (TPA). TriPleX ™

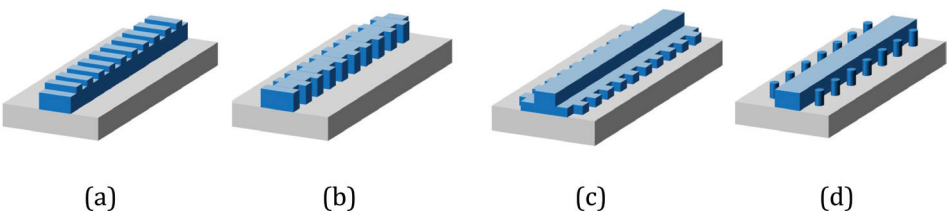


Figure 3.
Bragg grating waveguide with different spatial modulation: (a) waveguide height, (b) waveguide width, (c) slab width, and (d) cladding [51].

waveguide technology offers three main waveguides, including the box shape, and double-strip and single-strip waveguides [56]. The double-strip configuration provides a reasonable balance between the integration density and insertion loss [57]. The Si_3N_4 can be used for Kerr nonlinear photonics, and it has a large third-order nonlinear refractive index [58, 59].

The extreme polarization-dependent performance of the SOI devices demands polarization management schemes [43]. Note that there is the possibility of integrating the Si_3N_4 platform in the SOI platform or SOI waveguides in the Si_3N_4 platform to benefit from the functionalities of both [60]. Employing silicon oxynitride (SiON) core and silicon oxide cladding layers, the index contrast of the waveguide can be adjusted by varying the nitrogen-to-oxygen ratio. By this approach, the core index can be continuously varied for applications ranging from the low contrast demanded for fiber-matched processes to the high contrast required for compound semiconductor structures [61]. **Table 1** summarizes the waveguide parameters for different material categories including low/medium/high index contrast materials, $SiON_x$, III/V materials, and SOI [6].

Note that the unwanted ambient medium fluctuation may impact the performance of PICs due to the positive thermo-optic coefficient (TOC) of Si, SiO_2, and Si_3N_4, making it more critical to consider the temperature variations [62]. The effective index and TOC of the guided modes increase monolithically by increasing the waveguide width [63]. In the polymer-silica combination, the sign of the thermo-optic coefficients of the polymer and silica are opposite and the silicon substrate performs as a heat sink due to its high thermal conductivity [64].

Regarding the reconfiguration capability, the TOC of silicon nitride is much lower than silicon, which results in lower thermal sensitivity. Furthermore, the silicon nitrite-based waveguides suffer from larger dimensions due to moderate index contrast, but their phase errors are smaller for a specific fabrication error [63]. Since silicon nitrite-based reconfigurable devices have high power consumption due to their low thermo-optical coefficient, the suspended thermal phase shifters offer higher thermal efficiency and thus result in more compact devices [65].

Column	**1**	**2**	**3**	**4**	**5**	**6**
Characteristics	SiO_2 Low Δ	SiO_2 Low Medium Δ	SiO_2 High Δ	$SiON_x$	III/V	SOI
Index difference Δ (%) $\Delta = \frac{n_{core} - n_{clad}}{n_{core}}$	0.3	0.45	0.75	3.3	7.0 (46)	41 (46)
Core size (μm)	8 × 8	7 × 7	6 × 6	3 × 2	2.5 × 0.5(0.2 × 0.5)	0.2 × 0.5 0.3 × 0.3
Loss (dB/cm)	<0.01	0.02	0.04	0.1	2.5–3.5	1.8–2.0
Coupling loss (dB/point)	<0.1	0.1	0.4	3.7 (2)	5	6.8 (0.8)
Bending radius (mm)	25	15	5	0.8	0.25 (0.005)	0.002–0.005

Table 1.
Waveguide parameters for different material categories [6].

2.3 Bend waveguide design

Bend waveguides are key elements for the design of optical elements such as directional couplers, modulators, ring resonators, and interferometers. The curvature always attenuates the guided modes; thus, the geometry always suffers from radiation losses. Two main approaches to attack this problem are to either use decreasing curvatures or increase the mode confinement. These methods respectively lead to large devices and low fiber coupling efficiency [66]. Furthermore, the refractive index contrast of the core and clad impacts the amount of the bending [67]. Generally, by increasing the length of the waveguide and the refractive index contrast, the radiation losses can be reduced. Thus, in the photonic integrated circuits with compact waveguide routing, silicon on the insulator platform is preferred to silicon nitrite, since to reduce the mode mismatch loss in the bending regions in the low-loss Si_3N_4 waveguides, a large bend radius is required [68, 69]. Distortion of the transverse field profile, coupling between orthogonal polarization modes, and change of the phase constant are some other negative impacts of the curvatures on the waveguide performance, which can be controlled in carefully engineered devices.

To gain a vision of the curved structure's loss performance, a quasi-analytic technique based on the integration of a phenomenological absorption coefficient is proposed. The total loss is a function of the integral of the attenuation constant α at the local position s along the bend length *S* through [70]:

$$loss = \frac{10}{\log_e(10)} \int_0^S \alpha(s) ds \ (dB) \tag{2}$$

In general, the method applies to any shape function *f* that is continuous in its first derivative, such as sinusoidal bend [69, 70]. In this method, the bending loss per unit is represented by an exponential function [71]. The bending loss depends on the mode order in the multimode waveguides as [72]:

$$\alpha = K \exp(-cR), \text{where } c = \beta \left(\frac{2\Delta n_{eff}}{n_{eff}} \right)^{3/2} \tag{3}$$

The value of *K* depends on the refractive index of the core and cladding, and also the thickness of the waveguide; Δ_{neff} is the difference between the cladding refractive index and modal effective index n_{eff}, and *R* is the radius. Thus, as the bending radius or the mode order increases, the bending loss decreases [72].

Tapered cosine S-band may ensure better matching to the input array and connecting straight waveguides at their input ensures the single-mode propagation [30]. Moreover, by using spline bend waveguides, the losses can be shrunk with respect to circular bends, but the overall footprint remains in the same order. By introducing a small spline section at the strait-bend interface, the geometry can also be made compact. Creating the whole bend with multiple offset bends is another method for reducing the bending loss, which performs based on the reduction of the mode mismatch. The operational mechanism of this configuration is based on modulating the light path by the offset *δ*. Moreover, a 90° bend constructed by three sections, decorated by small offsets with each other, can be found in Ref. [73]. The problem with this technique is its performance sensitivity to the fabrication tolerance. Moreover, by optimizing the entire bend the losses can be minimized. This approach demands calculations for different bend angles and radii [74].

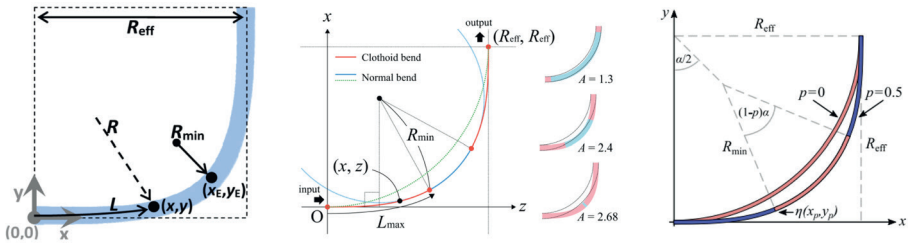


Figure 4.
Different types of Euler bends: (a) 90° normal Euler bend [75], (b) a normal bend terminated to the Euler bends in the output and input [76], and (c) partial Euler bend formed by an Euler bend terminated with normal bends [77] (reprinted with permission from aforementioned references © Optical Society of Optica Publishing Group).

The Euler bends have found applications in various integrated photonic devices. Since the eigenmode bases of straight and curved waveguides are different, the abrupt transition from infinite (in the straight section) to finite (in the bending section) curvature radius leads to spatial mode coupling. The bending-induced mode mismatch can be suppressed by the Euler bend [58]. The Euler curve, as shown in **Figure 4a**, also called the clothoid curve, has different radii of curvature in the bending path and results in low loss and low cross-talk [75]. Yet, the pure Euler curve is not an efficient way of bend waveguide design. To improve the performance, waveguide bends made up of the combinations of clothoid and normal curves are utilized for dense optical interconnect in photonic integrated circuits and they exhibit up to one order of magnitude reduction in the bending loss concerning the normal bend. The method is based on using the Euler curve on the input and output of the normal curve, where the ratio of the curves is achieved by optimization. This geometry can be found in **Figure 4b** [76]. In another method, the so-called partial Euler bend, the constant curvature waveguide input and output bands are connected to the Euler bend to balance the transition losses arising from a changing curvature and radiative losses inherent to a bend waveguide mode. The geometry of the partial Euler bend is shown in **Figure 4c** [77].

Finally, Intel, IBM, Luxtera, A*STAR, GLOBALFOUNDRIES, INPHOTEC, TowerJazz, LioniX, SMART Photonics, LIGENTEC, and Infinera are some of the companies developing photonic integrated technologies [78].

3. Conclusion

Optical waveguides are the building blocks of the photonic integrated circuits. By properly engineering the waveguide geometry, associated materials, the connecting curved waveguides, and fabrication method, it is feasible to reach low-loss devices. The designed low-loss structures may be efficiently used in conjunction with single-photon sources and detectors to conduct quantum computation. The final device potentially speeds up the calculation and gains the ability to perform calculations with some optimization strategies based on artificial intelligence.

Conflict of interest

The author declares no conflict of interest.

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